

CSE400 - Fundamentals of Probability in
Computing
Lecture 20: Linear Transformations and
Expectations of Multiple RVs

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March 19, 2025

Outline

- Statistical Representation of Random Vectors
- Mean Vector, Correlation, and Covariance Matrices
- Linear Transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of zero correlation and matrix factorization

1 Random Vector Representation

1.1 Vector Form of Multiple Random Variables

Random Vector

For random variables $X_1, X_2, \dots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- X is called a **random vector**.
- It provides a compact representation of multiple random variables.

2 Mean Vector

2.1 Expectation of a Random Vector

Mean Vector Definition

The mean vector of X is:

$$\mu_X = E[X]$$

- Represents the average value of the random vector.
- Each element corresponds to the mean of one random variable.

2.2 Example Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

3 Correlation Matrix

3.1 Correlation Between Multiple RVs

Correlation Matrix

$$R_{XX} = E[XX^T]$$

where $i, j \leq N$,

$$R_{XX}(i, j) = E[X_i X_j]$$

3.2 Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1 X_1] & E[X_1 X_2] & \cdots & E[X_1 X_N] \\ E[X_2 X_1] & E[X_2 X_2] & \cdots & E[X_2 X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_N X_1] & E[X_N X_2] & \cdots & E[X_N X_N] \end{bmatrix}$$

- R_{XX} is a symmetric matrix.

4 Covariance Matrix

4.1 Mathematical Definition

For a random vector X , the covariance matrix C_{XX} is:

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

4.2 Matrix Form in Practice

$$C_{XX} = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- Symmetry: $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$

5 Linear Transformation of Random Vectors

5.1 The Linear Model

Consider the transformation:

$$Y = AX + b$$

- A : Transformation matrix (scales and rotates the data)
- b : Constant vector (shifts/translated the center)
- X : Input random vector

For N dimensional vectors:

$$\begin{aligned} Y_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1 \\ Y_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2 \\ &\vdots \\ Y_N &= a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N \end{aligned}$$

6 Mean of Transformed Vector

6.1 Mean Transformation

For a linear transformation $Y = AX + b$:

$$\mu_Y = E[Y]$$

6.2 Derivation

Substitute $Y = AX + b$ into the expectation:

$$\mu_Y = E[AX + b]$$

Apply linearity:

$$E[AX] + E[b]$$

6.3 Result

$$\mu_Y = A\mu_X + b$$

7 Correlation Matrix Transformation

7.1 Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

7.2 Expansion of Terms

$$R_{YY} = E[AXX^T A^T + AXb^T + bX^T A^T + bb^T]$$

7.3 Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X^T]A^T + bb^T$$

7.4 Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

8 Covariance Matrix Transformation

8.1 Key Observation

The deviation from the mean is independent of the shift b :

$$Y - \mu_Y = (AX + b) - (A\mu_X + b) = A(X - \mu_X)$$